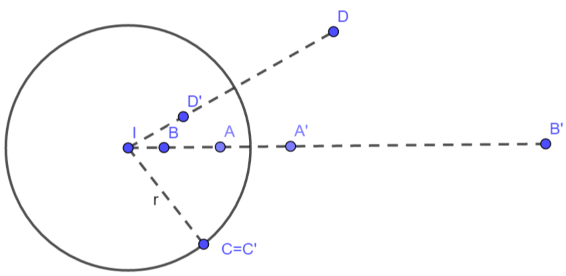
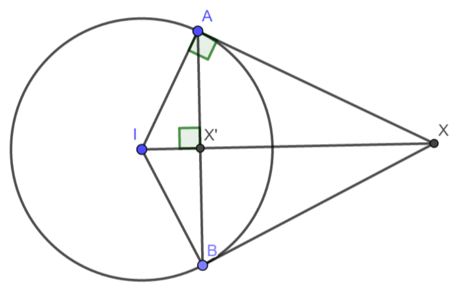
**IMO Training 2021**
**Inversion**
**22nd July, 2021**

Source: *107 Geometry Problems* by Titu Andreescu. The series of *106 Geometry Problems* and this book is the same source as most of the notes in level 1 by KKK.
A free source online would be the book *Euclidean Geometry in Mathematical Olympiads* by Evan Chen, where the chapter on inversion is legally free (`https://www.maa.org/sites/default/files/pdf/ebooks/pdf/EGMO_chapter8.pdf`).
Of course, another source would be the *Projective Geometry and Inversion* in our IMO training google drive under Additional Notes→有生之年系列→Intermediate Notes.

Given a circle $\omega$ with centre $I$ and radius $r > 0$, we define the image $X'$ of point $X$ under inversion about $\omega$ as the point on ray $IX$ such that

$$IX \cdot IX' = r^2.$$




Notice that

1. The points inside $\omega$ (i.e. $IX < r$) are mapped to the outside (i.e. $IX > r$); vice versa.
2. Inversion about the same circle twice is an identity mapping, i.e. $X'' = X$. In other words, $X'$ is the image of $X$ if and only if $X$ is the image of $X'$.
3. Points lying on $\omega$ are exactly those being invariant under inversion, i.e. $X' = X$.
4. Conventionally, the image of $I$ is $\infty$ and the image of $\infty$ is $I$.

The ruler-compass construction is as the right figure above, which follows from $\triangle IAX' \sim \triangle IXA$. In geogebra, it is the ”reflection about circle” action.
As we will see, the radius of inversion may often be chosen arbitrarily, therefore there would often be notions of ”inverting about a point”, meaning inverting about a circle with that point as centre and an arbitrary radius.

# 1 Properties

1. Let $I, X, Y$ be pairwise distinct non-collinear points. Denote by $X'$ and $Y'$ the images of $X$ and $Y$ under inversion about a circle with centre $I$ and radius $r > 0$.

    (a) $\triangle IXY \sim \triangle IY'X'$

    (b) $X, Y, Y', X'$ concyclic

    (c) (Angle chasing) $\angle XYI = \angle IX'Y'$

    (d) (Length chasing) $X'Y' = XY \cdot \dfrac{r^2}{IX \cdot IY}$ and $XY = X'Y' \cdot \dfrac{r^2}{IX' \cdot IY'}$

2. (Image of line) Denote by $\ell'$ the image of line $\ell$ under inversion about $I$.

    (a) If $\ell$ passes through $I$, then $\ell' = \ell$

    (b) If $\ell$ does not pass through $I$, $\ell'$ is a circle with center $O$ passing through $I$ such that $OI \perp \ell$.

3. (Image of circle) Denote by $\omega'$ the image of line $\omega$ with centre $O$ under inversion about a circle $\Gamma$ with centre $I$ and radius $r > 0$.

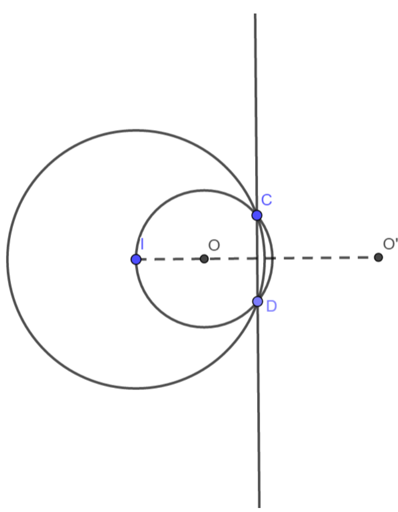
    (a) If $\omega$ passes through $I$, then $\omega'$ is a line perpendicular to $OI$.

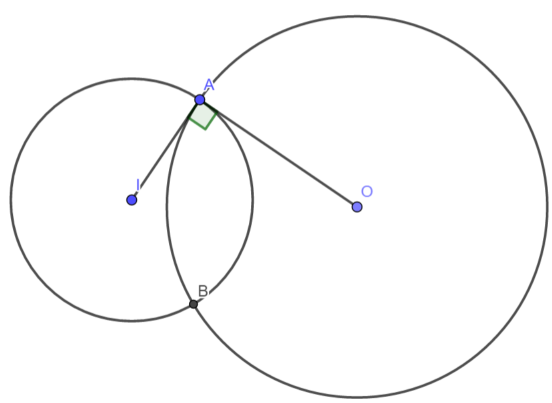
        i. $O'$ is the reflection of $I$ about line $\omega'$.

    (b) If $\omega$ does not pass through $I$, then $\omega'$ is a circle.

        i. In general, $O'$ is not the centre of $\omega'$. Still, the centre of $\omega'$ and the points $O, O'$ all lie on the same ray from $I$.

    (c) Suppose $\omega$ and $\Gamma$ intersect at points $A$ and $B$. Then $\omega' = \omega$ if and only if $\angle OAI = 90°$. In this case, we say $\omega$ and $\Gamma$ are *orthogonal*.



## 2 Equivalent Statements

A basic usage of inversion is to invert the whole question (including the statement to be proved) to obtain an equivalent problem, and sometimes the equivalent problem is easier to solve.

**Problem 1** (Ptolemy's Inequality)**.** Let $A, B, C, D$ be four points on the plane. Prove that

$$AB \cdot CD + AD \cdot BC \geq AC \cdot BD.$$

Show that equality case holds if and only if $A, B, C, D$ are either concyclic or collinear in this order.

**Problem 2** (USAMO 1993)**.** Let $ABCD$ be a convex quadrilateral whose diagonals intersect at right angle at $O$. Prove that the reflections of $O$ across lines $AB, BC, CD, DA$ are concyclic.

**Problem 3** (IMO 2003 Shortlist)**.** Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that $\Gamma_1, \Gamma_3$ are externally tangent at $P$, and $\Gamma_2, \Gamma_4$ are externally tangent at the same point $P$. Suppose that $\Gamma_1$ and $\Gamma_2$, $\Gamma_2$ and $\Gamma_3$, $\Gamma_3$ and $\Gamma_4$, $\Gamma_4$ and $\Gamma_1$ meet at $A, B, C, D$, respectively, and that all these points are different from $P$. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

**Problem 4** (IMO 1996)**.** Let $P$ be a point inside $\triangle ABC$ such that

$$\angle APB - \angle ACB = \angle APC - \angle ABC.$$

Let $D, E$ be the incentres of $\triangle APB$, $\triangle APC$ respectively. Show that the lines $AP, BD, CE$ meet at a point.

**Problem 5** (Romania TST 2008)**.** Let $ABC$ be a triangle with $\angle BAC < \angle ACB$. Let $D, E$ be points on the sides $AC$ and $AB$, respectively, such that $\angle ACB = \angle BED$. If $F$ lies in the interior of the quadrilateral $BCDE$ such that the circumcircle of $\triangle BCF$ is tangent to the circumcircle of $\triangle DEF$, and the circumcircle of $\triangle BEF$ is tangent to the circumcircle of $\triangle CDF$, prove that the points $A, C, E, F$ are concyclic.

**Problem 6** (IMO 2018)**.** A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point $X$ lies inside $ABCD$ so that $\angle XAB = \angle XCD$ and $\angle XBC = \angle XDA$. Prove that $\angle BXA + \angle DXC = 180°$.

## 3 Choice of inversion circle

A more sophisticated usage of inversion is to consider some effects of inversion without inverting the whole configuration. In this case, a wise choice of circle is crucial.

**Problem 7** (Iran 1995)**.** Let $M, N, P$ be the points where the incircle of scalene triangle $ABC$ touches its sides $BC, CA, AB$ respectively. Prove that the orthocentre of $\triangle MNP$, the incentre $I$ of $\triangle ABC$ and the circumcentre $O$ of $\triangle ABC$ are collinear.

**Problem 8** (Feuerbach's Theorem)**.** Show that the nine-point circle of a triangle is tangent to its incircle and three excircles.

**Problem 9** (IMO 2021)**.** Let $D$ be an interior point of the acute triangle $ABC$ with $AB > AC$ so that $\angle DAB = \angle CAD$. The point $E$ on the segment $AC$ satisfies $\angle ADE = \angle BCD$, the point $F$ on the segment $AB$ satisfies $\angle FDA = \angle DBC$, and the point $X$ on the line $AC$ satisfies $CX = BX$. Let $O_1$ and $O_2$ be the circumcentres of the triangles $ADC$ and $EXD$, respectively. Prove that the lines $BC$, $EF$, and $O_1O_2$ are concurrent.